Rhinitis

*By **Marvin P. Fried**, MD, Montefiore Medical Center, The University Hospital of Albert Einstein College of Medicine*

Reviewed/Revised Jun 2025 | Modified Jul 2025

Rhinitis is inflammation and swelling of the mucous membrane of the nose, characterized by a runny nose and stuffiness and usually caused by the common cold or a seasonal allergy.

- Colds and allergies are the most common causes of rhinitis.
- Symptoms of rhinitis include a runny nose and stuffiness; they may also include sneezing.
- Typically, the diagnosis is based on the symptoms.
- The various forms of rhinitis are treated in various ways, such as with antibiotics, antihistamines, surgery, allergy desensitization injections (sometimes called allergy shots or immunotherapy), and avoidance of irritants.

Rhinitis is classified as allergic or nonallergic. The cause of nonallergic rhinitis is usually a viral infection, although irritants can cause it. The nose is the most commonly infected part of the upper airways.

Rhinitis may be acute (short-lived) or chronic (long-standing). Acute rhinitis commonly results from viral infections but may also be a result of allergies, bacteria, or other causes. Chronic rhinitis usually occurs with chronic sinusitis (chronic rhinosinusitis).

Allergic Rhinitis

Allergic rhinitis is caused by a reaction of the body's immune system to an environmental trigger. The most common environmental triggers include dust, molds, pollens, grasses, trees, and animals. Both seasonal allergies and year-round allergies can cause allergic rhinitis.

Symptoms of allergic rhinitis include itching, sneezing, runny nose, stuffiness, and itchy, watery eyes. People may have headaches and swollen eyelids and also may cough and wheeze.

A doctor may diagnose allergic rhinitis based on a person's history of symptoms. Often, the person has a family history of allergies. More detailed information may be obtained from blood tests or skin testing.

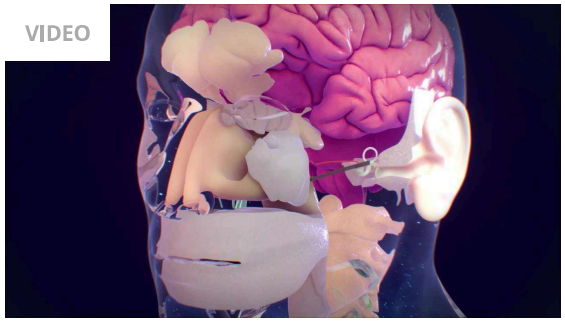
The following treatments can help avoid or treat symptoms of allergic rhinitis:

- **Avoiding the substance that triggers the allergy** prevents symptoms but is often not possible.
- **Nasal steroid sprays** (also sometimes referred to as glucocorticoid or corticosteroid sprays) decrease nasal inflammation from many sources and are relatively safe for long-term use.
- **Antihistamines** help prevent the allergic reaction and thus symptoms. Older antihistamines dry the mucous membrane of the nose, but many of them also cause sleepiness and other problems, especially in older adults. Some newer ones require a prescription but do not have as many of these side effects.
- A **saltwater solution** flushed through the nose via a squeeze bottle or a bulb syringe (nasal irrigation) or using a saltwater spray as needed also can help symptoms.
- **Desensitization injections** that contain small amounts of the substance that triggers the allergy (called desensitization injections or immunotherapy), or sometimes allergy shots) help to build long-term tolerance to specific environmental triggers, but they may take months or years to become fully effective. Sublingual immunotherapy (sometimes called allergy drops) are formulations designed to be administered under the tongue.

Antibiotics do not relieve the symptoms of allergic rhinitis.

The Nose

VIDEO



Acute viral rhinitis

Acute viral rhinitis can be caused by a variety of viruses, usually the common cold.

Symptoms consist of runny nose, sneezing, congestion, postnasal drip, cough, and a low-grade fever.

Stuffiness can be relieved by taking decongestants, such as oxymetazoline or phenylephrine as a nasal spray or pseudoephedrine by mouth. These medications, available over the counter, cause the blood vessels of the nasal mucous membrane to narrow (constrict). Nasal sprays should be used for only 3 days because after that period of time when the effects of the medications wear off, the mucous membrane often swells even more than before. This phenomenon is called rebound congestion.

Antihistamines taken by mouth or as a nasal spray sometimes help control a runny nose, but some cause drowsiness and most cause other problems, especially in older adults (see Aging and medications). Other medications such as mast-cell stabilizers, some of which also act as antihistamines, can control symptoms of allergic rhinitis.

Antibiotics are not effective for acute viral rhinitis.

Chronic rhinitis

Chronic rhinitis is usually an extension of rhinitis caused by inflammation or a viral infection. However, it also may rarely occur with diseases. These diseases include syphilis, tuberculosis, rhinoscleroma (a skin disease characterized by very hard, flattened tissues that first appear on the nose), rhinosporidiosis (an infection in the nose characterized by bleeding polyps), leishmaniasis, blastomycosis, histoplasmosis, and leprosy—all of which are characterized by the formation of inflamed lesions (granulomas) and the destruction of soft tissue, cartilage, and bone. Both low humidity and airborne irritants also can result in chronic rhinitis.

Chronic rhinitis causes nasal obstruction and, in severe cases, crusting, frequent bleeding, and thick, foul-smelling, pus-filled discharge from the nose. Sometimes partially healed tissues can cause hardening and distortion of the tissues inside the nose.

Decongestants may relieve symptoms. Any underlying infection requires a culture (examination of microorganisms grown from a sample of mucus to identify infection with bacteria or fungi) and appropriate treatment. If symptoms persist, biopsy (removal of a tissue sample for identification under a microscope) may be necessary to rule out cancer.

Atrophic rhinitis

Atrophic rhinitis is a form of chronic rhinitis in which the mucous membrane thins (atrophies) and hardens, causing the nasal passages to widen (dilate) and dry out. This atrophy often occurs in older adults. People who have granulomatosis with polyangiitis are also at risk. The cells normally found in the mucous membrane of the nose—cells that secrete mucus and have hairlike projections to move dirt particles out—are replaced by cells like those normally found in the skin. The disorder can also develop in people who had a significant amount of intranasal structures and mucous membranes removed during sinus surgery. A prolonged bacterial infection of the lining of the nose is also a factor.

Crusts form inside the nose, and an offensive odor develops. People may have recurring severe nosebleeds and can lose their sense of smell (anosmia).

Treatment is aimed at reducing the crusting, eliminating the odor, and reducing infections. Antibiotics, such as mupirocin ointment applied inside the nose, kill bacteria. Estrogens sprayed into the nose or taken by mouth and vitamins A and D taken by mouth may reduce crusting by promoting mucosal secretions. Surgery may sometimes be necessary to reduce the patency of the nasal cavities.

Vasomotor rhinitis

Vasomotor rhinitis is a form of chronic rhinitis. Nasal stuffiness and a runny nose—common allergic symptoms—occur when allergies do not seem to be present. Sneezing and nasal itching can occur but are less common than other nasal symptoms. In some people, the nose reacts strongly to irritants (such as dust and pollen), perfumes, pollution, or spicy foods. The disorder comes and goes and is worsened by dry and cold air. The swollen mucous membrane varies from bright red to purple. Sometimes, people also have slight inflammation of the sinuses. People do not have a pus-filled discharge or crusting.

Treatment of vasomotor rhinitis is by trial and error and may not always be satisfactory. If inflammation of the sinus is not severe, treatment is aimed at relieving symptoms. Avoiding smoke and irritants and using a humidified central heating system or vaporizer to increase humidity may be beneficial. Nasal steroid, anticholinergic, and antihistamine sprays sometimes help. Nasal decongestant sprays should not be used.

Rhinitis medicamentosa

Rhinitis medicamentosa also known as rebound congestion, is severe nasal congestion caused by the overuse (over 3 days of continuous use) of decongestant nasal sprays and drops (not from steroidal sprays). Treatment is by discontinuing the medication that is causing the condition and using a saline nasal spray. Steroid nasal sprays may also be used if needed.

MSD

Copyright © 2026 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. All rights reserved.